Python Practice Questions

1. Multiplication Table – School Mathematics Tool

A school teacher wants to create a small Python program to help students practice multiplication tables. Instead of manually writing the table every time, the teacher wants a program where a student enters any number and the program automatically generates its multiplication table from **1 to 10**.

Problem Statement

Write a Python program that accepts a number from the user and prints its multiplication table from 1 to 10.

Requirements

- Take one integer as input.
- Use a loop to generate the table.
- Print the result in the format:

5 x 1 = 5

5 x 2 = 10

...

5 x 10 = 50

Example

Input:

```
Enter a number: 7
```

Output:

```
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

Test Cases

Input	Expected Result
5	Table of 5 from 1–10
7	Table of 7 from 1–10
12	Table of 12 from 1–10
0	All results should be 0
-3	Table of -3 should be displayed

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2. Reverse a Number – ATM Security Verification

An ATM security system sometimes needs to process numbers in reverse order for verification purposes. For example, if a transaction reference number is 12345, the system should be able to generate 54321. You are asked to create a simple program that reverses a number **without converting it into a string**.

Problem Statement

Write a Python program to reverse a given integer using a loop.

Requirements

- Take an integer as input.
- Use mathematical operations and a loop.
- Do not convert the number into a string.
- Display the reversed number.

Example

Input:

```
Enter a number: 12345
```

Output:

```
Reversed Number: 54321
```

Test Cases

Input	Expected Output
12345	54321
9876	6789
1000	1
505	505
123	321

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3. Prime Numbers from 1 to 100 – Security Number Checker

A software company is developing a security application where prime numbers are used for certain internal calculations. The developer wants to generate all prime numbers between 1 and 100.

A prime number is a number greater than 1 that has exactly **two factors: 1 and itself**.

Problem Statement

Write a Python program to print all prime numbers between **1 and 100**.

Requirements

- Use loops.
- Check whether each number is prime.
- Print only prime numbers.
- Also display the total number of prime numbers found.

Example

Output:

```
Prime numbers between 1 and 100:
2
3
5
7
11
13
17
19
23
```

```
29
31
37
41
43
47
53
59
61
67
71
73
79
83
89
97

Total Prime Numbers: 25
```

Test Cases

Range	Expected
1–10	2, 3, 5, 7
1–20	2, 3, 5, 7, 11, 13, 17, 19
1–50	15 prime numbers
1–100	25 prime numbers

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4. Count Vowels and Consonants – Student Text Analyzer

A school is developing a simple text-analysis tool for students. The tool should analyze a sentence entered by the student and determine how many vowels and consonants are present.

For example, in:

Python Programming

the program should count the vowels and consonants separately.

Problem Statement

Write a Python program that accepts a string and counts:

- Number of vowels
- Number of consonants

Requirements

- Consider a, e, i, o, u as vowels.
- Handle uppercase and lowercase letters.
- Ignore spaces, numbers, and special characters.
- Use a loop.

Example

Input:

```
Enter a string: Python Programming
```

Output:

Vowels: 4
Consonants: 13

Test Cases

Input	Expected
hello	Vowels: 2, Consonants: 3
Python	Vowels: 1, Consonants: 5
Hello World	Vowels: 3, Consonants: 7
123 Python!	Vowels: 1, Consonants: 5
AEIOU	Vowels: 5, Consonants: 0

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5. Reverse a String Using a Function – Message Processing System

A messaging application wants to provide a small text utility that can reverse a user's message.

For example:

Python

should become:

nohtyP

The development team wants the operation to be reusable, so the reverse operation must be implemented inside a **function**.

Problem Statement

Create a function called reverse_string() that accepts a string and returns the reversed string.

Requirements

- Create a function.
- Accept a string as an argument.
- Return the reversed string.
- Call the function and display the result.

Example

Input:

Enter a string: Python

Output:

Reversed String: nohtyP

Test Cases

Input	Expected Output
Python	nohtyP
Hello	olleH
Codegnan	nangedoC
12345	54321
Hello World	dlroW olleH